

7.1.7. Let $A = \{1, 2, 3, 4, 5\}$, and define a function $F : \mathcal{P}(A) \rightarrow \mathbb{Z}$ as follows: For each set X in $\mathcal{P}(A)$,

$$F(X) = 0 \text{ if } X \text{ has an even number of elements}$$

$$= 1 \text{ if } X \text{ has an odd number of elements.}$$

Find the following:

- a. $F(\{1, 3, 4\})$
- b. $F(\emptyset)$
- c. $F(\{2, 3\})$
- d. $F(\{2, 3, 4, 5\})$

- a. $F(\{1, 3, 4\})$

$\{1, 3, 4\}$ has 3 elements, and 3 is odd.

$F(X) = 1$ if X has an odd number of elements.

Therefore, $F(\{1, 3, 4\}) = 1$

Solution: $F(\{1, 3, 4\}) = 1$

- b. $F(\emptyset)$

$\emptyset$ has 0 elements, and 0 is even.

$F(X) = 0$ if X has an even number of elements.

Therefore $F(\emptyset) = 0$

Solution: $F(\emptyset) = 0$

- c. $F(\{2, 3\})$

$\{2, 3\}$ has 2 elements, and 2 is even.

$F(X) = 0$ if X has an even number of elements.

Therefore $F(\{2, 3\}) = 0$

Solution: $F(\{2, 3\}) = 0$

- d. $F(\{2, 3, 4, 5\})$

$\{2, 3, 4, 5\}$ has 4 elements, and 4 is even.

$F(X) = 0$ if X has an even number of elements.

Therefore $F(\{2, 3, 4, 5\}) = 0$

Solution: $F(\{2, 3, 4, 5\}) = 0$